Raihan Fathurrahman

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SUMMARY

Bachelor of Computer Science graduate with comprehensive expertise in Full-Stack Development, Cloud Computing, and Data Science. Possesses hands-on experience in designing and executing high-impact software solutions, including a data processing automation system at PT Pertamina Hulu Rokan that drastically improved operational efficiency. Detail-oriented, adaptable, and ready to contribute to building efficient and scalable software architectures.

EXPERIENCE

IT Automation Developer – Drilling Engineering HO

PT Pertamina Hulu Rokan - Apprenticeship — August 2025—February 2026, Pekanbaru, Riau

- Conducted user needs assessment and translated engineering requirements into practical digital solutions.
- Engineered a comprehensive tool to automate the generation visualization of Stick Charts, Days vs. Depth (DvD) Charts and Basis of Design (BOD) documents.
- Supporting the engineering team in the Continuous Improvement Program (CIP) PHR in the creation of BLINC (Blind Liner for No Cement Plug) animations.
- Awardee Best Internship Participant Batch 7 PT Pertamina Hulu Rokan (PHR).

Support Staff

Office of Jon Erizal, Member of Indonesian House of Representatives 2019–2024 – Part time — Sep 2021—February 2024, Pekanbaru, Riau

- Assisted in organizing constituency visits (reses) and community outreach initiatives
- Managed social media content by creating posts, updating event activities, and ensuring consistent communication with constituents.
- Provided documentation through photography and videography, and assisted with basic administrative needs.
- Coordinated with local communities and stakeholders during field activities.

EDUCATION

Bachelor of Computer Science (S.Kom)

University of Muhammadiyah Riau • October 2021 – January 2025 • GPA 3.84/4.00

- Thesis Topic: Long Short-Term Memory (LSTM) for Fresh Fruit Bunch Price Forecasting
- Awardee Bidikmisi (KIP-K) Scholarship
- Completed studies in 3.3 years
- Awardee Best Graduate of Class of 28th with Cum Laude Predicate

TRAINING & CERTIFICATIONS

Bangkit Academy by Google, GoTo, Traveloka 2024

Cloud Computing Specialization • February 2024 – June 2024

- Completed intensive 900+ hour program focused on mastering Cloud Computing with Google Cloud Platform (GCP) services such as Compute Engine, App Engine, Cloud Functions, and Cloud Storage.
- Understand the concepts of DevOps, CI/CD, and efficient cloud infrastructure management.
- As team leader to develop a Capstone Project based on real technology solutions.
- Improve soft skills such as problem solving, communication, and team collaboration through industry training and mentoring sessions.

Google Cloud Computing Foundation Certificate

Google Cloud • April 2024

Google IT Support

Coursera • April 2024

Google Cyber Security

Coursera • July 2024

PROJECTS

Ray.Vis – AI Productivity Ecosystem

Personal Project • April 2026

Engineered a multi-module web application featuring AI-driven chat, automated voice-to-text transcription, financial tracking, media summarization, etc.

- Phase 1: Fullstack Web App (Deployed / V.1.0) - Design and develop a full productivity dashboard using the Next.js ecosystem, Tailwind CSS, and Firebase (Auth & Firestore NoSQL).
- Phase 2: Native Android Migration (In-Active Development) - Currently migrating the core web architecture into a native Android application using Kotlin and Jetpack Compose to achieve optimized mobile performance and native hardware access.

Automated Drilling Visualization

Internship Project – PT Pertamina Hulu Rokan • August 2025 – February 2026

- Automation tools with Macro VBA to automatically visualize Stick Chart, Days vs Depth Chart and Basis of Design based on offset well data from WellView.
- The project features a user-friendly filtering interface for well selection, streamlining reporting and accelerating decision making in upstream operations.
- Built a custom algorithm to automatically filter raw drilling data, distinguish complex well types and eliminate manual data-entry errors.
- Slashed the time required for data extraction and chart generation to under 1 minute per well scenario. Enhanced efficiency and reduced repetitive manual work for drilling engineers.

Time Series Forecasting of Fresh Fruit Bunch (FFB) Prices using LSTM

Thesis Project – Universitas Muhammadiyah Riau • September 2024 – January 2025

- Designed and developed a time series forecasting model to predict Fresh Fruit Bunch prices in Riau Province using the LSTM algorithm based on 10 years of weekly data (2014–2024).
- Performed data preprocessing including noise reduction, feature selection and data splitting with 2024 reserved for testing to simulate real-world forecasting.
- Demonstrated the model's capability to capture temporal patterns and provide decision-support insights for stakeholders in the palm oil industry, particularly in Riau Province, Indonesia.

Ocular Disease Detection using InceptionV3

Research Project • December 2024 – January 2025

- Developed a deep learning model for eye disease classification using transfer learning with the InceptionV3 architecture, leveraging pre-trained weights to improve accuracy and reduce training time.
- Utilized 1559 dataset of eye images, applying preprocessing and augmentation techniques to enhance model performance and generalization.
- Achieved promising classification results across multiple eye conditions and compiled the research findings into a scientific journal manuscript.

EyeTify Mobile App

Capstone Project – Bangkit Academy • April 2024 – June 2024

- Led a team in developing deep learning-based mobile application for early detection of eye diseases through image classification, integrating machine learning models with the backend to enhance job matching.
- Designed and built scalable APIs using Express.js a nd Firestore, deployed o n Cloud Run and App Engine for availability.
- Optimized cloud resources, managing Cloud Storage for secure data handling and ensuring cost-effective allocation with Google Cloud Platform.

PUBLICATION

Pendekatan Transfer Learning untuk Klasifikasi Penyakit Mata Menggunakan Citra dengan CNN InceptionV3

Jurnal Computer Science and Information Technology (CoSciTech) • SINTA 4 • May 2025 • Access Link

- Developed a Deep Learning-based eye disease classification model using the InceptionV3 architecture to detect eight different types of eye diseases.
- Successful implementation of Transfer Learning techniques improved training efficiency and model accuracy on a limited medical dataset (1,559 images).
- Performed data preprocessing and dataset splitting (80% training, 20% validation) to ensure good model generalization and prevent overfitting.
- Analyzed model performance results to demonstrate the effectiveness of the CNN architecture in assisting in the automatic early detection of vision disorders.

AWARDS

- Peserta Magang Kerja Terbaik PT Pertamina Hulu Rokan (2026) - Company
- Wisudawan Terbaik Universitas Muhammadiyah Riau Angkatan XXVIII dengan Predikat Cumlaude (2025) - University

SKILLS

Technical Skills:

- Languages: PHP, Python, JavaScript
- Frameworks/Libraries: Next.js, React, Node.js, Laravel, RESTful APIs
- Database: MySQL, SQL Server
- DevOps: Google Cloud (GCP), Docker, CI/CD, Git, Firebase
- Data Analytics: PowerBI, Tableau, Macro VBA
- Tools/Productivity: MsOffice, Virtual Desktop Infrastructure

Soft Skills: Problem Solving, Research, Team Work, Time Management.
Language: English (Advanced), Indonesian (Fluent)